# starlab

## _travelers guide
## _for deep space exploration

| 3 | 4 | 15 |
|---|---|---|
| textures | favorites | scales |

**[ infinite ]** press  **[ infinite ]** hold  › then  noon = 12 o'clock  cw / ccw = clockwise / counter

time-warped reverberator · reverb, delay, karplus-strong string, shimmer, glimmer, lfo



## 01 buttons · LEDs

sl_01

| | |
|---|---|
| **[ infinite ]** | freezes what's going into the reverb · new sound keeps passing through |
| **[ favorite ]** | recalls the selected preset (09) |
| **[ regen ]** | ● shimmer inside the reverb, it climbs · off = shimmer on the input only |
| **[ high band ]** | ● glimmer lifts the high harmonics · off = the low ones |
| **[ echo on ]** | ● the delay joins the wet output · off = it's a pre-delay |
| **[ tap / trig ]** | tap a tempo · karplus: quick press plucks, hold bows |
| **[ low pass ]** | high damp becomes a 24 db resonant low pass on the output |

## 02 reverb

sl_02

| | |
|---|---|
| **[ texture ]** | **sparse** granular · **dense** plate-like · **diffuse** slow, swelling wash |
| **[ size / pitch ]** | cw = bigger room, lower pitch · ±1 octave (⅛× to 2× size) · retunes the tail |
| **[ decay ]** | how long the tail rings · long settings also feel the filters |
| **[ dry ]** | analog, never digitized · none at min · unity at noon · +3 db at max |
| **[ wet ]** | reverb (and echo) level |
| **[ input gain ]** | into the reverb · LED ● clean › ● soft clip · ● in gate muted |

## 05 harmonics

sl_05

| | |
|---|---|
| **[ shimmer ]** | pitch-shifted reverb · off fully ccw |
| **[ regen ] + [ shimmer ]** | **interval**: −1 oct to +1 oct · ±10 cent detune either side of noon |
| **[ glimmer ]** | brings out harmonics already in your sound · off fully ccw |
| **[ interval ]** cv | shimmer interval in half steps, ±1 octave |

## __fig.06 : lfo shapes

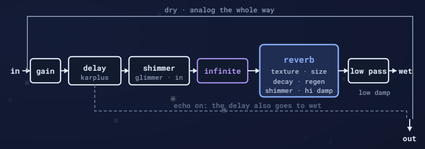
**[ shape ]**




env follows how loud you play · depth sets its direction, off at noon

## __fig.01 : signal path

left to right




echo off = the delay is a pre-delay · infinite holds the input, the reverb keeps running
regen on = shimmer inside the reverb · off = shimmer on the way in

## 03 delay

sl_03 · delay / karplus

| | |
|---|---|
| **[ delay / tune ]** | 0–1.5 s, finer at the short end · with echo off and feedback low it's a pre-delay |
| **[ feedback ]** | more repeats, or resonance at short times |
| **[ echo on ]** | sends the delay to wet beside the reverb |
| **[ tap / trig ]** | tap the time · a clock in the jack turns delay / tune into a ratio (fig.02) |
| delay only | **[ decay ]** at min + **[ echo on ]** |
| longest | 7.5 s: a 3 s clock with delay / tune on 5/2 |

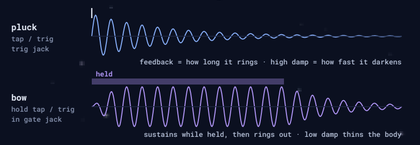
## 04 karplus-strong

sl_04 · a string voice

| | |
|---|---|
| **[ echo on ] + [ delay ]** cw | until tap / trig turns ● = string mode · ccw until it blinks ● = delay again |
| **[ delay / tune ]** | tunes the string over 4 octaves · with size cv patched it becomes a 12-semitone fine tune |
| **[ tap / trig ]** | quick = pluck · hold = bow · trig jack plucks, in gate jack bows |
| **[ feedback ]** | how long the string rings |
| **[ echo on ]** | adds the dry string to wet · off = only the reverberated string |
| pitch cv | **[ size / pitch ]** jack, 0–4 v, 1 v/oct |

## __fig.04 : pluck and bow

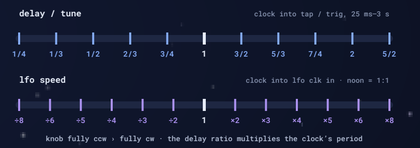
karplus-strong mode




pitch: delay / tune (4 oct) or size / pitch cv 0–4 v · echo on adds the dry string

## __fig.02 : clocked knobs

what each position means




knob fully ccw › fully cw · the delay ratio multiplies the clock's period

## 07 lfo

sl_07

| | |
|---|---|
| **[ lfo target ]** | **delay** chorus, flange, vibrato (echo on for more) · **pitch** size / pitch · **filter** high damp (low pass on for sweeps) |
| **[ speed ]** | 15 s to 15 hz · env: the decay time |
| **[ shape ]** | triangle · square · ramp · saw · random · env (follows your input) |
| **[ depth ]** | off fully ccw · env: off at noon, ccw negative, cw positive |
| **[ lfo clk in ]** | locks to a clock · speed picks the ratio (fig.02) |
| **[ ext lfo in ]** | your own modulation to the target · depth becomes its attenuverter, off at noon |

## 08 pitch quantize

sl_08 · **[ infinite ]**

| | |
|---|---|
| **[ infinite ]** | until it blinks ● = size / pitch steps through a scale · again until ● = off |
| **[ infinite ] + [ size ]** | pick one of 15 scales · the 4 center buttons show it (fig.05) |
| **[ low pass ] + [ size ]** | portamento between notes · min = none |

## __fig.05 : the 15 scales

**[ infinite ] + [ size ]**




## 09 favorites

sl_09 · four presets

| | |
|---|---|
| **[ favorite ]** | recall the selected one |
| **[ favorite ] + [ 1–4 ]** | choose · the current one shows ● |
| **[ favorite ]** long | until ● › **[ favorite ]** saves here, or a center button saves there |
| **[ favorite ]** cv | toggles between the knobs and the preset · dip 1 on = spillover between them |

## 10 cv inputs

sl_10

**continuous, ±5 v** — the knob is the offset

| | |
|---|---|
| **[ size / pitch ]** | ±2 v, 1 v/oct: ±2 octaves · karplus 0–4 v |
| also | wet · decay · shimmer · interval · glimmer · high damp · feedback · speed · depth · ext lfo |

**triggers, 0–5 v rising edge**

| | |
|---|---|
| **[ infinite ] [ echo on ]** | toggle · latching by default |
| **[ in gate ]** | mutes the input while high · karplus: bows |
| **[ favorite ] [ clear ]** | knobs ‹› preset · empties the reverb |
| **[ tap / trig ] [ lfo clk in ]** | clocks, 25 ms–3 s · karplus: pluck |

## 06 filter

sl_06 · into the reverb tank

| | |
|---|---|
| **[ low damp ]** | cuts lows in the output and the regenerating core |
| **[ high damp ]** | cuts highs in the regenerating tail |
| **[ low pass ]** | high damp becomes a 24 db / oct low pass on the output |
| **[ low pass ] + [ high damp ]** | resonance · cw = a sharper peak |

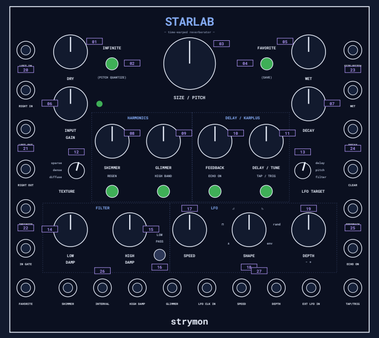
## 11 the module

sl_11 · 2021

| | |
|---|---|
| size | 28 hp · 41 mm deep · 210 mA on each rail · 96 khz, 32-bit float · analog dry path |

## __fig.03 : front panel

01–19 = controls · 20–27 = jacks




| | |
|---|---|
| [ 01 ] | **__dry :** analog dry |
| [ 02 ] | **__infinite :** freeze · hold = quantize |
| [ 03 ] | **__size / pitch :** ±1 oct |
| [ 04 ] | **__favorite :** presets · hold = save |
| [ 05 ] | **__wet :** effect level |
| [ 06 ] | **__input gain :** drive · LED |
| [ 07 ] | **__decay :** tail length |
| [ 08 ] | **__shimmer · regen :** pitch shift |
| [ 09 ] | **__glimmer · band :** harmonics |
| [ 10 ] | **__feedback · echo :** repeats · to wet |
| [ 11 ] | **__delay · tap :** time · tune · pluck |
| [ 12 ] | **__texture :** sparse · dense · diffuse |
| [ 13 ] | **__lfo target :** delay · pitch · filter |
| [ 14 ] | **__low damp :** lows out |
| [ 15 ] | **__high damp :** highs out |
| [ 16 ] | **__low pass :** 24 db lp · hold = q |
| [ 17 ] | **__speed :** 15 s–15 hz |
| [ 18 ] | **__shape :** six waves |
| [ 19 ] | **__depth :** amount · attenuverter |
| [ 20 ] | **__ins :** left = mono |
| [ 21 ] | **__outs :** stereo |
| [ 22 ] | **__infinite · gate · fav :** triggers |
| [ 23 ] | **__size · wet · decay :** cv |
| [ 24 ] | **__clear :** empty the tail |
| [ 25 ] | **__fb · echo · tap :** cv · clock |
| [ 26 ] | **__shimmer → glimmer :** cv |
| [ 27 ] | **__lfo jacks :** clk · speed · depth · ext |

## _where things hide

no menus · sl_12

### held buttons

- **infinite** pitch quantize · + size = scale
- **regen + shimmer** shimmer interval
- **echo on + delay** karplus ‹› delay
- **low pass + …** high damp = q · size = glide · gain = tracking
- **favorite** + 1–4 = pick · long = save

### at power on

- **hold regen** infinite cv latch ‹› momentary
- **hold high band** in gate cv
- **hold echo on** echo on cv
- **dip 1 (back)** favorite spillover

### LED codes

| | |
|---|---|
| **input gain** | ● clean ● soft clip ● in gate muted |
| **infinite** | ● blink = quantize on ● blink = off |
| **tap / trig** | ● karplus ● blink = delay |
| **favorite** | ● chosen preset ● flash = save |